

# Quantum and classical discrete time crystals

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# Table of Contents

1 Introduction

2 Time-Crystalline Phase

3 Floquet Hamiltonian Systems

- Faraday waves
- Floquet Hamiltonians and emergent symmetries
- Ergodicity

# Table of Contents

- 4 Closed Quantum System
  - Quantum Floquet phases
  - MBL DTC
  - Prethermal DTC
  - Periodically driven BEC
- 5 Open Quantum and Classical System
  - Activated DTC
  - Ergodicity in open systems
  - Probabilistic cellular automata
  - Open Hamiltonian and quantum dynamics
- 6 Conclusions

# Landau formalism

Landau-Ginzburg theory: spontaneous symmetry breaking

e.g. BCS  $\phi \rightarrow e^{i\lambda}\phi$  as  $U(1)$  symmetry

$$V(\phi) = b_4|\nabla\phi|^4 + b_2|\nabla\phi|^2 + a_2|\phi|^2 + a_4|\phi|^4 = H = -L \quad (1)$$

with  $a_4, b_4 > 0$  for stability

$a_2 < 0$ : phase rotation symmetry;  $b_2 < 0$ : spatial symmetry

# Time-translation symmetry

$$L_{\text{kin}} = c_4 |\partial_t \phi|^4 + c_2 |\partial_t \phi|^2 \rightarrow H_{\text{kin}} = 3c_4 |\partial_t \phi|^4 + c_2 |\partial_t \phi|^2 \quad (2)$$

$c_2 < 0, c_4 > 0$ : time-translation symmetry breaking

called **spontaneous time-translation symmetry breaking (S<sub>T</sub>B)**

But something subtle

- stationary at energy minima
- $\partial_t \phi$  is not a single-valued function of  $p$  so as Hamiltonian

## Watanabe-Oshikawa's no-go theorem (2015)

In an equilibrium state of a time-independent Hamiltonian with not-too-long interactions, time-crystals (defined by correlation functions) doesn't exist.

Realize in the limit of large quantum numbers (semi-classical limit)

### Discrete $S\tau B$

$$H(t + \tau) = H(t), \quad \langle O(t + \tau) \rangle \neq \langle O(t) \rangle \quad (3)$$

but subharmonic responses

$$\langle O(t + n\tau) \rangle = \langle O(t) \rangle, \quad \omega = \omega_D/n \quad (4)$$

Similar to crystals.

# Ergodicity breaking

Spontaneous symmetry breaking is a special case of ergodicity breaking.

**$S\tau B$** : it is a form of ergodicity breaking in which a generic ensemble of initial conditions exhibits persistent oscillations with a temporal phase shift that remembers the initial condition.

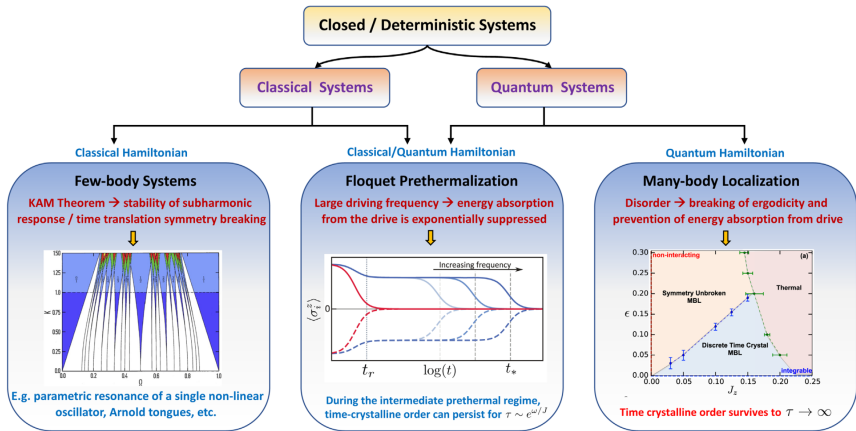
If this  $S\tau B$  is stable to perturbations, the system is a "discrete time crystal". **The possibility of perfect  $S\tau B$  in the presence of a drive, dissipation, and noise remains an open question.**

# Interesting types of DTC

Large quantum number limit

- 1 MBL time crystal: not generate entropy,  $\tau \sim$  system size
- 2 Prethermal time crystal: driven frequency  $\omega_D \rightarrow \infty$
- 3 Activated time crystal:  $T \rightarrow 0$
- 4 Driven BEC time crystal: particle density  $N \rightarrow \infty$



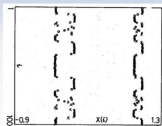


## Open / Stochastic Systems

Classical Langevin / Quantum Lindblad

Temperature,  $T = 0$ Temperature,  $T > 0$ 

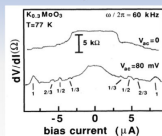
## Dissipative Dynamical Systems

Damping / friction  $\rightarrow$  perturbations to either the state or the dynamics decay rapidly

E.g. coupled logistic maps as a realization of a dissipative many-body time crystal

Time crystalline order survives to  $T \rightarrow \infty$ 

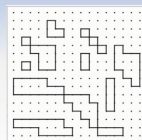
## Thermally Activated

Low temperature  $\rightarrow$  dissipation dominates

E.g. AC-driven charge density wave materials, parametrically driven coupled non-linear pendula

Time crystalline order persists for  $T \sim e^{\Delta/k_B T}$ 

## Toom / Gacs Models

"Error correction"  $\rightarrow$  breaking of ergodicity

E.g. cellular automaton models (Gacs model in 1D, Toom model in 2D) as many-body time crystals

Time crystalline order survives to  $T \rightarrow \infty$

# What is DTC?

From oscillations: state  $x$ , discrete-time update rule  $\Phi : x \rightarrow \Phi(x)$  and  $x(t) = \Phi^{(t)}(x(0))$  at time step  $t$  (both classical and quantum).

The time independence of  $\Phi$ : discrete time-translation symmetry

## ★ m-fold time-translation symmetry breaking

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn)}(x)) \neq \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn+p)}(x)) \quad (5)$$

where  $0 < p < m$ . So it can restore  $m$  initial conditions.  $Z_m = Z/mZ$ .

E.g.  $\Phi(x) = -x$  as  $Z_2$

$\Phi^{(m)}$  depends on the initial condition: **asymptotic periodicity** for classical many-body systems.

- Infinite timescale  $\tau$  or finite timescale  $\tau$  which diverges with a control parameter
- Observable  $O$  oscillates on average, **not**  $x$ .

Equation (5)  $\sim$  evolution microstates  $\rho(x)$  (classical and quantum)

### Stability

$S\tau B$  is a property of a **dynamical phase** of matter if it is robust to any small locality-preserving perturbation of either the initial condition  $x$  or the dynamics  $\Phi$ .

E.g.  $\Phi(x) = -x(1 - \epsilon)$  thus  $\tau \sim \epsilon^{-1}$ , so it's not DTC.

Different perturbation  $\rightarrow$  different  $S\tau B$  phase

## Some examples

Without dissipation (classical zero-temperature limit of open system)

- Undriven oscillator:  $H = p^2/2m + (\omega_0^2/2m)q^2 + (\epsilon/4)q^4$   
 $T \sim \epsilon \rightarrow$  not DTC, ensemble will dephase
- AC Josephson effect:  $H = -E_J \cos \phi + 2eVn + n^2/2C$   
 $\phi$ : superconducting phase difference across the junction,  
 $n$ : Cooper pair number,  
 $E_J$ : Josephson energy (one pair tunneling),  
 $C$ : junction capacitance,  
 $V$ : voltage difference across the junction  
 $\tau = \sqrt{C/E_J}$  may be large  $\rightarrow$  'DTC' in practice from external drive.

- Superfluids ( $\mu = 0$ ) and an XY magnet in a perpendicular field  $U(1)$  protect  $S\tau B$ , but not DTC
- Logistic map, e.g.  $f(x) = -x(1 + a - x^2)$ : one-body sample coupled map lattice  $\Phi : x_i \rightarrow f(x_i) + v(x_{i-1}, x_i, x_{i+1})$  called cellular automata with collective subharmonic responses.

**Stable** to smooth perturbations.

- van der Pol oscillator (closed limit cycle), limit-cycle oscillators, ...  $\neq S\tau B$  because of its continuous-time dynamics.

# Faraday-wave instabilities in shaken surface waves

Standing-wave modes are forced via parametric resonance. (subharmonics)

## Mathieu equation

Linearizing the incompressible Euler equations for surface waves

$$\ddot{q}_k = - \left[ \omega_k^2 + \delta_k \cos(\omega_D t) \right] q_k \quad (6)$$

where  $q_k$ : Fourier component of the surface height  $q$

$\delta_k \propto$  driving amplitude

One solution: exponentially growing solution  $\sim e^{t\Gamma} \cos(t\omega_D/2 + \theta)$  when  $\omega_D \sim 2\omega_k$

**Nonlinearity stabilizes  $\mathbf{S\tau B}$ .** (Hamiltonian by Euler equation)

Long time? Nonlinear eq.: high- $k$  modes acts as effectively noisy.

# Floquet Hamiltonians and emergent symmetries

Example: shaken nonlinear single pendulum (specific  $k$  mode)

$$H(q, p, t) = \frac{p^2}{2} + \frac{\omega_0^2}{2} [1 + \delta \cos(\omega_D t)] q^2 + \frac{\epsilon}{4} q^4 \quad (7)$$

**Nonlinearity stabilizes  $S\tau B$ .**

KAM  $\rightarrow$  stable  $S\tau B$ ; Include interacting  $\rightarrow$  unstable (ergodicity?)

Floquet Hamiltonian for closed classical/quantum  $S\tau B$



# Rotating frame

Near a period-doubled solution:

$$q(t) - ip(t) = \sqrt{2P(t)} e^{i[\omega_D t/2 + Q(t)]} \quad (8)$$

where  $Q(t), P(t)$ : slowly varying  $\sim \omega_D$ . Just like rotating-wave approximation.

Give **time-dependent canonical transformation**  $\mathcal{K}_0(t)$ :  $(q, p) \rightarrow (Q, P)$  called **rotating frame**

Subharmonic:  $\mathcal{K}_0(mT) = \mathcal{K}_0(0)$  with  $\omega_{\text{sub}} = \omega_{\text{driven}}/m$ ,  $m \in \mathbb{Z}$

★ **Period-doubling: fixed points in rotating frame**

# Floquet Hamiltonian

## First transformation

$H(q, p, t) \rightarrow H(Q, P, t)$ : natural frequencies are off resonant with  $\omega_D$

## Second transformation: **Magnus expansion**

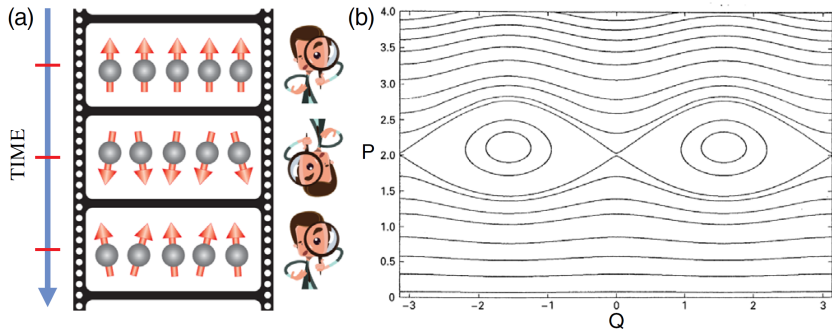
To generate time-indepnedent  $H_F$

$\mathcal{K}(t) = \mathcal{K}_{\text{Magnus}}(t) \circ \mathcal{K}_0(t)$  still with  $\mathcal{K}(mT) = \mathcal{K}(0)$  (**converge?**)

$\mathcal{K}(t) : (q, p) \rightarrow (Q, P)$ : Driven system governed by  $H(q, p, t) \Leftrightarrow$

Equilibrium system governed by  $H_F(Q, P)$  in rotating frame.

(crypto-equilibrium)



# Floquet Hamiltonian

## Classical

Magnus expansion converge?

- True DTC: well-defined  $H_F$
- Prethermal DTC:  $H(Q, P, t) = H_{eff}(Q, P) + V(t)$ , left  $V(t)$  (the exponentially small)

# Floquet Hamiltonian

## Quantum

$$U(t_1, t_0) = \mathcal{T} \exp \left( -i \int_{t_0}^{t_1} H(t) dt \right) = \mathcal{K}^{-1}(t_1) e^{-i(t_1 - t_0) H_F} \mathcal{K}(t_0) \quad (9)$$

where  $H_F$  suitably local.

$\mathcal{K}(t)$  is called micromotion  $\sim \omega_D/m$ ;  $H_F \sim$  slow part.

One step (stroboscopic dynamics):  $U(t_1, t_0) \rightarrow U_F = U(T, 0)$ ;  $\Psi \rightarrow U_F$ ,  
 $x = |\psi\rangle$

# Internal symmetry breaking

- Trivial:  $\mathcal{K}(T) = \mathcal{K}(0)$ , no  $S\tau B$  because it's forbidden in equilibrium.
- Nontrivial:  $\mathcal{K}(mT) = \mathcal{K}(0)$ , subharmonic

$$X = \mathcal{K}(0)\mathcal{K}^{-1}(T) \quad (10)$$

E.g. for single pendulum,  $X : (Q, P) \rightarrow (Q + \pi, P)$

More general: under Magnus expansion

$$X = \mathcal{K}(nT)\mathcal{K}^{-1}(nT + T), \quad X^m = I \quad (11)$$

is independent with  $n$ .

# Internal symmetry breaking

$$U(t_1, t_0) = \mathcal{K}^{-1}(t_1) e^{-i(t_1 - t_0)H_F} \mathcal{K}(t_0) \quad (12)$$

with time-translation symmetry  $U(T, 0) = U(2T, T)$ , we get

$$X^{-1} e^{-iTH_F} X = e^{-iTH_F} \quad (13)$$

**X is an internal  $Z_m$  symmetry of the Floquet Hamiltonian (twisted time translation)**

# Emergent symmetry

How about  $S\tau B$ ? (Construction)

**The dynamics of  $H_F$  spontaneously break the  $X$  symmetry.**

That is, **the state space breaks up into  $m$  superselection sectors (permuted by  $X$  but not  $H_F$ )**

For different initial conditions  $y_{j+1} = X y_j$ , local observables

$$\bar{O}(y_j) \equiv \frac{1}{\tau} \sum_{n=0}^{\tau} O(y_j(nT)) \neq \bar{O}(y_{j+1}) \quad (14)$$

then transform to lab frame  $x_j = \mathcal{K}^{-1} y_j$ .



# Emergent symmetry

$m$ -fold  $S\tau B \longleftrightarrow Z_m$ -spontaneous symmetry breaking

A general principle of all **closed Hamiltonian many-body time crystals**:

In a **Hamiltonian** time crystal, the discrete time-translation symmetry of  $H(t)$  manifests as a **spontaneously broken internal**  $Z_m$  symmetry of the Floquet Hamiltonian  $H_F$ .

For Landau formalism,  $S\tau B: H \rightarrow H_F$

# Spectral perspective

**Equivalent definition (quantum):** DTC is a phase of matter in which the eigenstates of  $U_F$  are necessarily cat states, i.e., superpositions of macroscopically distinct states.

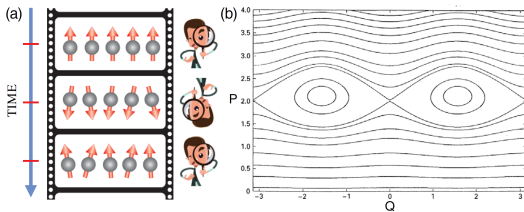
## Floquet eigenstates

$$U_F |\epsilon_F\rangle = e^{-iT\epsilon_F} |\epsilon_F\rangle \quad (15)$$

Driving:  $\epsilon_F \equiv \epsilon_F + \omega_D$ , same state

# Application to many-body parametric resonance

- In this 0D case, the  $S\tau B$  arises because KAM stability allows the quasienergy to split up the  $(Q, P)$  space into disconnected basins, rather than because of a collective phenomena.
- In 0D there is long-range order in time but not in space, and as such one may prefer to **not** consider this case a time crystal.



# Application to many-body parametric resonance

**Why the quantum version of a parametrically driven nonlinear oscillator will not feature infinitely long-lived  $S_{\tau}B$ ?**

While a quantum Floquet Hamiltonian with a  $Z_2$  symmetry may exist, for any finite barrier height there will be **quantum tunneling** between the two minima at a rate  $1/\tau$ . This leads to a **unique  $Z_2$ -symmetric steady state**, and the subharmonic response will have a correlation time of  $\tau$ .

# Application to many-body parametric resonance

## Many-body setting (nearest-neighbor interaction)

Parametrically driven 1D Frenkel-Kontorova model

$$H(q, p, t) = \sum_i \left[ \frac{p_i^2}{2} + \frac{\omega_0^2}{2} [1 + \delta \cos(\omega_D t)] q_i^2 + \frac{\epsilon}{4} q_i^4 \right] - g \sum_{\langle i, j \rangle} (q_i - q_j)^2 \quad (16)$$

Macroscopic object composed of atoms

- If  $q_i, p_i = q, p$  for all  $i$ , stable period doubling
- Interaction term may remain time-dependent after transformations, so we cannot appeal to the conservation of  $H_F$  to ensure stability.  
because of **Ergodicity**

# Definition of ergodicity in Hamiltonian systems

Hamiltonian system: **"measure perserving"** by Liouville theoerm

Definition: ergodicity in invertible measure-preserving system

The system is said to be ergodic **if** (equivalent)

- the only finite-volume subset  $A$  that is invariant under the discrete-time update rule  $\Phi$  is the entire phase space.
- the orbit of any volume  $\bigcup_m \Phi^{(m)}(A)$  eventually fills all of phase space.

# Definition of ergodicity in Hamiltonian systems

How about **full ergodicity**

- 1 with a time-indepnedent  $H_F(Q, P)$ ?

However they are incompatible because quasienergy  $H_F$  breaks phase space into  $\Phi$ -invariant quasienergy contours, while ergodicity exist in each contour.

- 2 with a time-depnedent  $H_F(Q, P, t)$  (bounded state space)?

The many-body system is to be ergodic in the thermodynamic limit.

- 3 unbounded state space? Heating up.

# Time-dependent Hamiltonian with bounded state space

## Ergodic hypothesis with Floquet setting

Time-independent  $\mathcal{H} = H(t) + \pi_t$ , where  $(t, \pi_t)$  is a canonical pair on the cylinder  $t \sim t + T$

If  $\mathcal{H}$  is ergodic on an energy shell, then the stroboscopic dynamics of  $H(t)$  are fully ergodic.



# Birkhoff's ergodic theorem (2012)

$S\tau B$  requires  $\Phi^{(m)}$  to be non-ergodic, that is, ergodicity breaking.

- Interaction: weak; Driving: strong.

Slow **Floquet heating**: Prethermal time crystals

Timescale (explore all of phase space) can be large  $\rightarrow$  practical " $S\tau B$ ".

$H(Q, P, t) = H_{eff}(Q, P) + V(t)$ , the quasienergy  $\langle H_{eff} \rangle$  change slowly due to  $V(t)$ .

# Examples about ergodicity

- Faraday waves with **viscosity**

- ① converts the high- $k$  oscillations of surface into heat.
- ② breaks measure-preserving, analogous to the coupled map lattices (true  $S\tau B$ ).

## Fluctuation-dissipation theorem: "Open" time crystals

At any finite temperature (energy density), a microscopic bath invariably generates noise in addition to viscosity.

- Noise and viscosity

# Where is stable $S\tau B$ ?

Stable  $S\tau B$  is possible **in**

- ① Dissipative dynamical systems (coupled map lattices, limit cycle of a van derPol oscillator)
- ② Few-body classical systems (parametric resonator)
- ③ **Many-body localization**

and **not in**

- ① Closed classical systems (except fine-tuned)

# Floquet-MBL systems

Many-body localization ( $\tau \sim e^{W/\Omega}$ ) is a phenomenon that is believed to occur in

- 1 1D systems with strong quenched disorder and short range interactions.
- 2 1D periodically driven systems with strong disorder for appropriate driving parameters (**Floquet-MBL**, no proof)

# Floquet-MBL systems

## Example: topology in Floquet system

Free fermion models: Majorana chain, driven between topological superconducting and trivial insulating states.

$$\text{Floquet operator } U_F = \exp \left( ic \sum_j i\gamma_j^A \gamma_j^B \right) \exp \left( iJ \sum_j i\gamma_j^B \gamma_{j+1}^B \right) \quad (17)$$

when  $c$  near  $\pi/2$ , called "Floquet Majorana fermions" with quasienergy  $\pi$ .

Jordan-Wigner transformation: driven Majorana chain  $\rightarrow$  driven Ising model.

Stable: strong disorder against interactions.

# Floquet-MBL systems

## DTC as a phase

Discrete time crystals are a stable phase of matter in Floquet-MBL systems with only time-translation symmetry.

DTC in closed systems must satisfy (5)

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn)}(x)) \neq \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn+p)}(x)) \quad (18)$$

no matter that if other symmetries are violated.

# Floquet-MBL systems

Open questions:

- In  $d > 1$  or **long-range interactions** for undriven MBL?
- Mathematical proof for the stability of MBL  $S_{\tau}B$  in a Floquet system **in any dimension?**

Another choice: Floquet prethermalization

## Example: Period-doubled DTC

Disordered spin-1/2 chain

$$H(t) = \begin{cases} H_1, & 0 < t < t_1; \\ H_2, & t_1 < t < T (T = t_1 + t_2). \end{cases} \quad (19)$$

where

$$H_1 = - \sum_{\langle ij \rangle} J_{ij} \sigma_i^z \sigma_j^z - \sum_i (h_i^z \sigma_i^z + h_i^y \sigma_i^y + h_i^x \sigma_i^x) \quad (20)$$

$$H_2 = g \sum_i \sigma_i^x \quad (21)$$

with disordered  $J_{ij}, h_i^x, h_i^y, h_i^z$ . When  $t_2 = \frac{\pi}{2}g$ , called  $\pi$ -pulse (flip spin, period-doubled response).



How about stability?

- Decoupled, unstable
- $J_{ij} \neq 0$ , MBL, stable (weak, **period- $nT$**  perturbation)

**Period- $nT$**  perturbation isn't necessary in **open systems**.

Experiments: trapped atomic ions, spins in solid, superconducting qubits...

Sycamore processor

$$U_F = \exp\left(-\frac{i}{2} \sum_i h_i \sigma_i^z\right) \exp\left(-\frac{i}{4} \sum_i J_i \sigma_i^z \sigma_{i+1}^z\right) \exp\left(-\frac{i}{2} \pi g \sum_i \sigma_i^x\right) \quad (22)$$

# Floquet prethermalization

**Floquet prethermalization is a generic feature of driven systems in the high-frequency regime:** heating timescale  $t_* \sim e^{\omega_D/J}$

- MBL lifetime  $\tau \sim e^L (L \rightarrow \infty)$ ; Time-crystalline order  $\tau \sim e^{\omega_D/J}$
- Fast dynamics:  $H_{eff}$ ; Slow dynamics: frozen ( $\omega_D$  and  $J$  are independent)

**Physical intuition (classical and quantum):**

At  $\omega_D \gg J$  (where  $J$  is the local energy scale of the many-body system), the system must undergo local rearrangements ( $N \sim \omega_D/J$ , Poisson distribution) in order to absorb a single unit of energy from the drive.

# Floquet prethermalization

That is

- 1 After  $t_*$ , absorb energy from the drive
- 2  $H_{eff}$  govern stroboscopic Floquet dynamics  $\rightarrow$  **emergent symmetry**  
(protected by discrete time-translation symmetry of the drive)

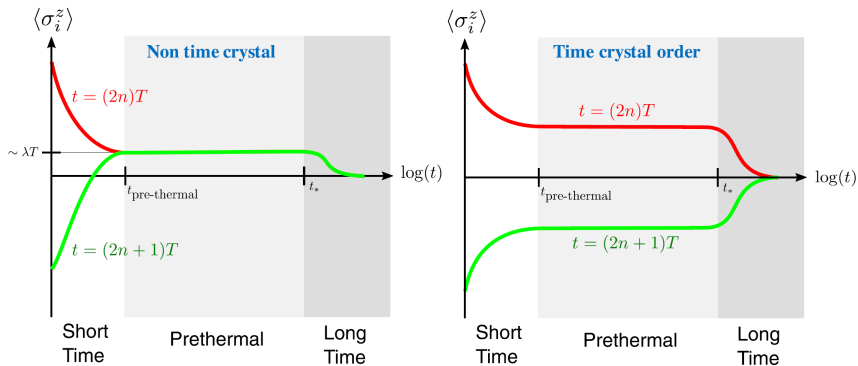
## Proven

If a many-body system prethermalizes to a state that spontaneously breaks the emergent symmetry of  $H_{eff}$ , it will also exhibit time-crystalline order, corresponding to a subharmonic oscillation between the different symmetry sectors.

# Floquet prethermalization

How to realize prethermal discrete time crystals (PDTC)?

- 1 Emergent symmetry in prethermal  $H_{eff}$  (define phase of matter)
- 2  $H_{eff}$  host a symmetry-broken phase with respect to the emergent symmetry. (Interaction range and dimensionality by Landau-Peierls / Mermin-Wagner)
- 3 Initial state has a low energy density ( $H_{eff}$ ) so that it can run during the prethermal regime,  $T < T_c$ .



## PDTC in a 1D trapped ion chain

At leading order in the Floquet-Magnus expansion: disorderless, long-range, mixed-field Ising model

$$H_{eff} = \sum_{\langle ij \rangle} J_{ij} \sigma_i^x \sigma_j^x + B_y \sum_i \sigma_i^y \quad (23)$$

with  $\pi$  pulse. If the system exhibit time-crystalline order, the pseudoequilibrium state breaks the emergent Ising symmetry.

In 1D, only for long-range interactions.

Symmetry-unbroken/Symmetry-broken state:  $M(t)$  decay quickly/period doubling

# Periodically driven BEC

BEC bouncing off an oscillating mirror:

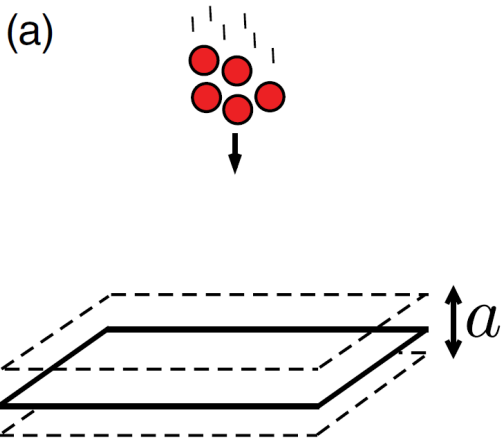
2:1 subharmonic response, quantum tunneling results in a finite lifetime  $\tau$  ( $\tau_{BEC}$  longer).

$$H = \int_0^\infty dz \left\{ \Psi^\dagger(z) \left[ -\frac{1}{2} \partial_z^2 + \left( 1 + \frac{a}{g} \cos(\omega_D t) \right) |z| \right] \Psi(z) - \frac{g_0}{4} |\Psi(z)|^4 \right\} \quad (24)$$

Approximations

- ①  $N \rightarrow \infty$  while  $g_0 N$  hold fixed, periodically driven Gross-Pitaevskii equation.
- ② Project interaction to two Floquet eigenstates,  $\tau \propto e^{\alpha N}$ .

or time-dependent Bogoliubov theory + the truncated Wigner approximation. Floquet heating?





# Open Systems

- Markov process  $\rho(x', t + T) = \int \Phi(x'|x)\rho(x, t)dx$
- Quantum channel  $\hat{\rho}(t + T) = \Phi[\hat{\rho}(t)]$

Bath: absorbed entropy v.s. phase slips

Fokker-Planck equation:  $\partial_t \rho = \mathcal{L}[\rho]$

# Langevin dynamics

Langevin force:  $F_L(t) = -\eta\dot{q} + f(t)$  with  $\langle f(t)f(t') \rangle = 2\eta T\delta(t - t')$ , where  $T$  is the temperature of bath.

$$\dot{p} = -\partial_q H(t) + F_L(t) \quad (25)$$

time-independent  $H$  (detailed balance)  $\rightarrow$  canonical ensemble, no  $S\tau B$

## Damped Mathieu equation

$$\ddot{q}_k = - \left[ \omega_k^2 + \delta_k \cos(\omega_D t) \right] q_k - \gamma_k \dot{q}_k \quad (26)$$

with  $H(P, Q, t) = H_{eff}(P, Q) + V(t)$ .

- Dissipation ( $T = 0$ ), stable limit cycle.
- Noise ( $T > 0$ ), finite autocorrelation time, lifetime  $\tau \sim e^{\Delta/k_B T}$ , called activated time crystals.

# 1D coupled array of pendula

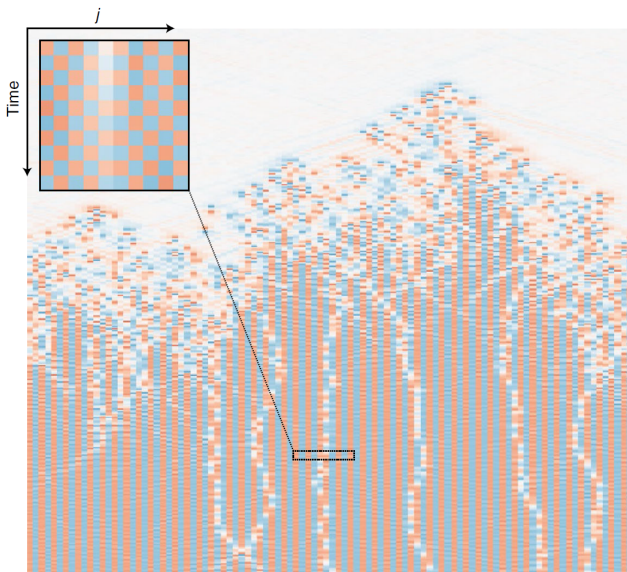
$$H_{eff} = \sum_i \left[ \frac{(P_i - P_*)^2}{2} + a \cos(2Q_i) \right] + \frac{\tilde{g}}{2} \sum_{\langle ij \rangle} (Q_i - Q_j)^2 \quad (27)$$

where gas of domain wall behaves Brownian motion, nucleating or annihilating in pairs.

$$\langle q_i(t = nT) q_i(t = 0) \rangle \sim (-1)^n e^{-t/\tau}, \quad \tau \sim e^{\Delta/k_B T} \quad (28)$$

$d \geq 2$ : equilibrium Ising symmetry breaking,  $\tau \rightarrow \infty$  when  $T < T_c$ ? No, rotation results forces.

$$V \simeq a \cos(2Q) + \eta P_* Q \quad (29)$$



# Three regimes of time-crystalline behavior

- ① Closed case ( $\eta = T = 0$ ): prethermal,  $\tau \sim e^{\omega_D/J}$
- ② Open case ( $\eta > 0, T > 0$ ): activated,  $\tau \sim e^{\Delta/k_B T}$
- ③ Purely dissipative case ( $\eta > 0, T = 0$ ): **ture**  $S\tau B$ ,  $\tau \rightarrow \infty$

Experiments of activated time crystal: shaken pendulum, ac-driven charge density waves (CDWs), fractional Shapiro steps in Josephson junction arrays.

# Definition of ergodicity in open systems

Invariant measure  $\rho_s$ :  $\Phi[\rho_s] = \rho_s$ , steady-state.

## Ergodicity

A stochastic system is said to be ergodic if the following two properties hold:

- ① the dynamics have a unique steady state  $\rho_s$
- ② the long-time behavior of any initial state relaxes to this steady state:  

$$\lim_{t \rightarrow \infty} \rho(t) = \rho_s$$

Similar to the closed case

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn)}(x)) \neq \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \sum_{n=1}^{\tau} O(\Phi^{(mn+p)}(x)) \quad (30)$$

## Ergodicity breaking: asymptotic periodicity

$$\lim_{t \rightarrow \infty} \rho(t) = \sum_{p=1}^m \alpha_p \rho_{p+1} \quad (31)$$

where

$$\Phi[\rho_p] = \rho_p, \quad \rho_{p+m} = \rho_p \quad (32)$$

can still define  $\rho_s = \frac{1}{m} \sum_p \rho_p$  (special case).

**Can a locally interacting stochastic system generically break ergodicity?**



# Kinetic Ising model

Detailed balance,  $\rho_s = e^{-\beta H} / Z$

- $T > 0$ :  $\rho_s$  is the unique steady state with a finite relaxation time, ergodic
- $h = 0, T < T_c$ : two steady state, not ergodic (need "fine-tuned" parameters)

How to break? Multibody interaction / Power-law interaction of sufficiently slow falloff

# From kinetic Ising model to CA

More general local Markov updates  $\Phi(\sigma'|\sigma)$  without  $H$ :

**probabilistic cellular automata (PCA)** (Interacting particle systems in the continuous-time generalization)

## CA

Deterministic cellular automata (CA):  $\sigma \rightarrow \Phi_{CA}(\sigma)$ . With Markovian error rate  $\epsilon$  to flips spin randomly, we get a perturbation of the deterministic CA:  $\Phi = \Phi_\epsilon \circ \Phi_{CA}$ .

Note: PCA isn't Markovian necessarily. CA are **Turing complete**.

## Time crystal from CA

Are there CA that remain nonergodic for generic small perturbations  $\Phi_\epsilon$ ?

And if so, might such a CA be the basis for S $\tau$ B? **Yes!!!**

# Bennett's answer to the second question (1990)

Assume:

- ① a local, nonergodic CA:  $\Phi_{CA}$ , with  $m$ -distinct steady state holding  $Z_m$  (for simplicity).
- ② the nonergodicity is stable to generic (and possibly  $Z_m$  breaking) errors.

"Rotating" CA

$$\Phi_{(2\pi/m)CA} = G_{2\pi/m} \circ \Phi_{CA} \Rightarrow \Phi_{CA}^{(m)} = \Phi_{(2\pi/m)CA}^{(m)} \quad (33)$$

if  $\Phi_{CA}$  is proven to be stably nonergodic,  $\Phi_{(2\pi/m)CA}$  exhibits the asymptotic periodicity of a time crystal, cycling through the  $m$  fixed steady states of  $\Phi_{CA}$ .

**$S\tau B$  does not depend on any further internal symmetries.**

## Toom's answer to the first question (1980)

Does a local and generically nonergodic PCA exist?

### Toom CA

The Toom CA is a 2D binary-spin model with a simple "northeast corner" (NEC) majority rule (choose most one)

$$\sigma_{x,y}(t+1) = \text{maj}(\sigma_{x,y}, \sigma_{x+1,y}, \sigma_{x,y+1}) \quad (34)$$

Note: No Hamiltonian, No detailed balance

The NEC rule will cause the island to shrink linearly from the NE direction inward, far faster from the Ising model.

# Toom's answer to the first question (1980)

## Toom PCA

Toom is Toom CA with biased errors. The bias is

$$b = \frac{\epsilon_p - \epsilon_q}{\epsilon_p + \epsilon_q} \quad (35)$$

where  $\epsilon_p$ : flip up;  $\epsilon_q$ : flip down.

$b \neq 1/2$  breaks Ising symmetry **but** there is a finite volume in  $\{\epsilon_p, \epsilon_q\}$  space remain two steady state, which robust to any sufficiently small perturbation (proven).

## Absolute stability

# Toom's answer to the first question (1980)

## $\pi$ -Toom model

Antimajority NEC rule (choose least one)

$$\sigma_{x,y}(t+1) = \text{antimaj}(\sigma_{x,y}, \sigma_{x+1,y}, \sigma_{x,y+1}) \quad (36)$$

The  $\pi$ -Toom model is a time crystal, which is stable to arbitrary perturbations, called full error-correcting capability. (2020)

"type-G" phase transition. (2000)

## Toom's answer to the first question (1980)

- Toom-like models exist in all dimensions  $d > 1$  simply by stacking the 2D version.
- Gács models allow stable ergodicity breaking in 1D PCA.
- PCA support absolutely stable  $S\tau B$  of any period  $m \geq 2$  in all dimensions  $d \geq 1$ .

Why  $\pi$ -Ising model can exhibit in  $m = 2$  but not in  $m > 2$ ?

Chirality.(2019)

How about activated time-crystals?

# Return to microscopic view

From Toom PCA to Langevin / Linblad equation?

- The  $\pi$ -Toom model can be embedded in classical Langevin / quantum Linblad dynamics.(2021)
- Toom-like stability was proved for a continuous-time version of a PCA with discrete state space. (2001)

**Is error-corrected computing physically realizable in this Universe?**



# Summary

Infinitely long-lived time crystals:

- $S\tau B$  that is stable to time-translation-invariant perturbations might occur in periodically driven closed 1D quantum systems in the presence of strong disorder

By strongly localizing the degrees of freedom

- $S\tau B$  that is stable to arbitrary perturbations can occur in periodically driven  $D \geq 1$  open systems, such as the  $\pi$ -Toom model, whose dynamics effectively implements a form of error correction

By actively shrinking potentially ergodicity-generating fluctuations

# Summary

Exponentially long-lived time crystals:

- prethermal time crystals in closed systems (both classical and quantum)
- activated time crystals in open systems

# Open questions

1. How does the transition from exponentially to infinitely long-lived time crystals precisely occur?
  - Sharp phase transition between the prethermal DTC and the Floquet-MBL DTC, similar as between the activated DTC to the Toom-type time crystal?
  - Exponentially long-lived time crystals's lifetimes: 1. prefactor diverges; 2. the effective energy scale  $1/J$  or  $\Delta$  diverges.

# Open questions

2. Whether is a direct transition possible from a Floquet-MBL discrete time crystal to a  $\pi$ -Toom or Gács-type time crystal (or something continuously connected to it)?

- Where the system goes from closed to open.
- Floquet-MBL DTC: perfect quantum computer;  $\pi$ -Toom time crystal: fault-tolerant quantum computer

Since the former could be simulated by the latter, one might also logically expect them to be separated by no more than a second-order phase transition

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